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Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2025

Code: 101305 | ***Time: 03 Hours***

Course: B.Tech | ***Subject: Fluid Mechanics*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) With increase in temperature, viscosity of liquids
- (i) Increases
- (ii) Decreases
- (iii) Remains same
- (iv) Becomes zero

(b) Newton's law of viscosity relates
- (i) Pressure & velocity
- (ii) Shear stress & velocity gradient
- (iii) Density & temperature
- (iv) Force & area

(c) Pascal's law states that pressure applied to fluid is
- (i) Lost
- (ii) Reduced
- (iii) Transmitted equally
- (iv) Increased

(d) Steady flow means
- (i) Velocity changes with time
- (ii) Velocity remains constant with time
- (iii) Pressure changes
- (iv) Density changes

(e) Laminar flow occurs at
- (i) High Reynolds number
- (ii) Low Reynolds number
- (iii) Zero velocity
- (iv) High pressure

(f) Ideal fluid has
- (i) High viscosity
- (ii) Zero viscosity
- (iii) High density
- (iv) Variable pressure

(g) Continuity equation represents
- (i) Energy conservation
- (ii) Mass conservation
- (iii) Momentum conservation
- (iv) Pressure variation

(h) Euler's equation is based on
- (i) Energy conservation
- (ii) Momentum conservation
- (iii) Mass conservation
- (iv) Heat transfer

(i) Mach number is ratio of
- (i) Velocity to sound speed
- (ii) Pressure to density
- (iii) Force to area
- (iv) Density to viscosity

(j) Hagen-Poiseuille law is valid for
- (i) Turbulent flow
- (ii) Laminar flow
- (iii) Compressible flow
- (iv) Ideal flow

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Q.2 Solve both questions : **[14]**

(a) Define the following : (i) Mass density (ii) Specific Gravity (iii) Dynamic Viscosity (iv) Capillarity (v) Vapour Pressure.

(b) The dynamic viscosity of oil used for lubrication between a shaft and sleeve is 8 poise. The shaft is of diameter 0.4 m and rotates at 190 rpm. Calculate the power lost in the bearing for a sleeve length of 90mm. The thickness of the oil film is 1.5mm.

Q.3 Solve both questions : **[14]**

(a) What is manometer? Distinguish between U-tube differential manometer and inverted U-tube differential manometers.

(b) Explain the procedure of finding hydrostatic force on curved surface.

Q.4 Solve both questions : **[14]**

(a) Distinguish between (i) Study and Unsteady flow (ii) Uniform and non-uniform flow (iii) compressible and incompressible.

(b) If for a two-dimensional potential flow, the velocity potential is given by Ø = x(2y - 1). Determine the velocity at the point P (4, 5). Also determine the value of stream function Ψ at the point P.

Q.5 Solve both questions : **[14]**

(a) Define the following terms: Velocity potential function, stream function, equipotential line and flow net.

(b) The water is flowing through a pipe having diameters 20cm and 10cm at section 1 and 2 respectively. The rate of flow through pipe is 35 litres/sec. The section 1 is 6m above datum and section 2 is 4m above datum. If the pressure at section 1 is 39.24N/cm^2. Find the intensity of pressure at section 2.

Q.6 Solve both questions : **[14]**

(a) Drive an expression for head loss due to sudden enlargement in a pipe flow.

(b) Prove that the maximum velocity in a circular pipe for viscous flow is equal to two times the average velocity of flow.

Q.7 Solve both questions : **[14]**

(a) At a sudden enlargement of a water main from 240 mm to 480 mm diameter, the hydraulic gradient raises by 10mm. Estimate the rate of flow.

(b) The rate of flow of water through a horizontal pipe is 0.25m^3/s. The diameter of the pipe which is 200 mm is suddenly enlarged to 400 mm. The pressure intensity in the smaller pipe is 11.772 N/cm^2. Determine: (i) Loss of head due to sudden enlargement (ii) Pressure intensity in the large pipe (iii) Power loss due to enlargement.

Q.8 Solve both questions : **[14]**

(a) Explain the different type of hydraulic similarity that must exists between a prototype and its model.

(b) Drive on the basis of dimensional analysis suitable parameter to present the thrust development by a propeller. Assume that the thrust P depends upon the angular velocity ω, speed of advance V, diameter D, dynamic viscosity μ, mass density ρ, elasticity of the fluid medium which can be denoted by the speed of sound in the medium of C.

Q.9 Write short note on any two of the following: **[7 x 2 = 14]**

(a) Define the term velocity potential function, stream function, and establish relation between them

(b) State Buckingham's π theorem

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(c) Obtain an expression for continuity equation for three dimensional flow

(d) What is capillary? Obtain an expression for capillary rise of a liquid in a glass tube

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